Gen4 Application Note

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1. Introduction

This document describes the new features of Flux Vector in Dragon8 and HVLP (and HDSRDS).

It assumes prior knowledge of Flux Vector setup in Gen4 and prior knowledge of basic configuration of the aforementioned controllers.

For simplicity, the term 'Genus' will be used to refer to the Dragon8/HVLP/Controllers as this is the of the common software platform these products share.

The flux vector scheme used in Genus is effectively the same algorithm used in Gen4.

There are some differences to simplify the setup process:

- As ScWiz is NOT initially supported, Genus features some auto-calculation features for mutual inductance (Lm) and rotor/stator resistances (rr, rs). These can be used as starter parameters for Applications Engineers.
- Slew rates for Id and Iq are no longer present as they could cause instability.
- Speed dependent field weakening is no longer present as ls/ls' can already achieve this.
- Maximum slip is no longer present (as this breaks the flux vector control).
- Current control loops run at 2x the PWM frequency, so nominally 16kHz. This is 4x faster than Gen4 and should improve system stability.

Flux vector setup is the same in all Genus products.

The only difference is that the AB encoder circuitry may require an internal pull-up to be set on Dragon8, whereas HVLP is differential-pair 5V input, so this is not applicable.

Please ensure firmware and DVT are up-to-date.

2. Quick reference

Some key info contained in this document:

To set HVLP/Dragon8/HDSRDS firmware in Flux Vector mode

- 0x4639,0 = 0x0023

New parameters for debug (viewable on vehicle interface trace)

- 0x4602,52: User specified magnetizing inductance (Lm)
- 0x4602,53: Controller estimated magnetizing inductance (Lm)
- 0x4602,55: Controller estimated stator resistance (Rs)
- 0x4602,56: Controller estimated rotor resistance (Rr)

As with Gen4, the user must populate the object dictionary with motor parameters. The estimated parameters are visual aids to help with commissioning.

To build up a recording of Lm saturation vs Id current,

- Set value of Lls & Llr in 0x4641 if known. Otherwise set to near-zero (e.g. 0x0001)
- run the dyno at speed
- enable controller with zero torque
- adjust id_min (0x4641, 3) to set Id through a range of currents.
- observe estimated Lm (0x4602,53) at changes with Id current.

Once an Lm cold value (0x4641,9) and the saturation lookup table objects are populated(0x4614), the User Specified Magnetizing Inductance will show what value of Lm is currently in-use. Note: Leakages need to be taken away from estimated value of Lm if you've set Lls & Llr to 0x0001(D8\HVLP) or using Gen4. (As a general rule - 10%)

To set the controller into signal-inject mode,

- sdo_wnx 1 0x4653 0 0x0020

Lm, Rs and Rr will update every 4 seconds. Ensure controller is enabled and at-speed on a dyno.

Characterisation spreadsheet:

- N:\UK\Engineering\Documents\Motor Control\Vector Spreadsheets

3. Firmware DLDX and Starter DCF

Genus requires two DLDX files for firmware. An MCP and IOP .dldx pair should be loaded into the motor and IO processor nodes.

There is NO specific firmware for Flux Vector, simply obtain up-to-date firmware for your Genus product.

Firmware contains algorithms for ALL supported motor types. The user selects between motor algorithms using object 0x4639.

For flux vector 0x4639,0 = 0x0023

The best way to start a Flux-Vector setup is to obtain a .dcf of a similar motor. This is often sufficient to obtain rotation and allow further tuning of parameters.

Please consult UK applications for a starter DCF.

4. Existing and New Motor Control Parameters

Genus uses the same key motor control parameters as Gen4.

These are:

- 0x4641 AC Motor data
- 0x4610 Motor Flux map
- 0x4614 Im/Lm saturation map.

Encoder configuration objects have moved and are described on page 7.

While all items in 0x4641 are writeable, those used by Flux Vector are:

2	Maximum Stator Current (Is_max)
3	Minimum Magnetizing Current (Im_min)
5	Number of Pole Pairs (np)
6	Stator Resistance (Rs)
8	Rotor Resistance (Rr)
9	Magnetizing Inductance (Lm)
10	Stator Leakage Inductance (Lls)
11	Rotor Leakage Inductance (Llr)
12	Nominal battery voltage
13	Current control proportional gain (Kp)
15	Current control integral gain (Ki)
25	Frequency/Mod index control Kp
26	Frequency/Mod index control Ki
30	Max drive mod index
31	Max brake mod index
32	Ls/Ls'
33	Voltage Control Kp
33	D-axis current controller proportional gain
34	D-axis current controller integral gain
45	Iq and Id Kp scaling factor for low-frequency PWM.
46	Iq and Id Ki scaling factor for low-frequency PWM.

All parameters are present in Gen4 Flux Vector and operate in the same manner. The exceptions are:

- 45 and 46 scale the PI gains for variable frequency PWM. Their use is not described here (as they are not specific to Flux Vector) and sensible defaults should be present in the starter .dcf.

Some parameters have been deleted since Gen4 as they were not deemed useful:

- Maximum slip (as it breaks the flux vector algorithm – Ls/Ls' and the power limit map should be used to limit the slip)
- Im rated (speed dependent field weakening is not present)

There are no configuration bits specific to flux-vector. The code operates in the same manner as Gen4 with the saturating Lm bug fixed.

To aid with tuning, the following new debug items are present in 0x4602 and should be included on Vehicle interface traces:

- 0x4602,52: User specified magnetizing inductance (Lm)
  - This shows the Lm in use – based on value in 0x4641 and saturation table in 0x4614
- 0x4602,53: Controller estimated magnetizing inductance (Lm)
  - An estimate of Lm based on observed currents/voltages. Always active.
- 0x4602,55: Controller estimated stator resistance (Rs)
  - An estimate of stator resistance.
  - Updates only when in signal inject mode.
- 0x4602,56: Controller estimated rotor resistance (Rr)
  - An estimate of rotor resistance.
  - Updates only when in signal inject mode.

As with Gen4, the user must populate all motor parameters and there is no 'self-learning'. The new debug items are to aid the user when performing a characterisation.

5. Encoder and sequence

Genus Flux-Vector, can theoretically operate with Resolver, sin-cos, UVW or AB with however this document describes AB as it is most common.

- Ensure AB is selected 0x4631,1 0x03.
- For Dragon8 ensure encoder pull-up is selected if required (0x4631,11 bit 0x0004),
- Ensure the correct PPR is set in 0x4635,3 (AB settings).
- Ensure motor pole pairs in 0x4641 is set correctly.
- Encoder supply is set in 0x4631,13 as with Gen4.
    - This is for HVLP only, but note that HVLP is 5V differential AB input to the controller.

Rotating the motor should show a changing encoder angle in 0x4631 and no encoder faults.

If a starter .dcf is present, then either on a dyno or on a vehicle, for a small torque demand, see if motor rotation occurs. If not, firstly, ensure the controller is enabling correctly (are slip, Id and Iq, Ud and Uq changing?)

If the controller is enabling, then change the encoder sequence and retry. A reversed sequence can typically results in a little burst of rotation before the motor stops.

To change the sequence, either swap the A,B wires, change the order of the M1,M2,M3 terminals or set the invert angle bit in 0x4631,11 bit zero.

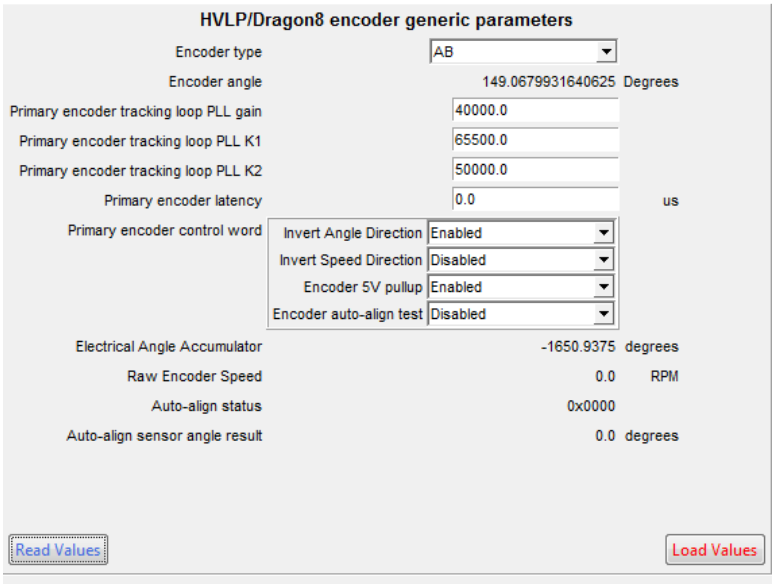


Figure 1: 0x4631 – Typical settings and with invert angle bit and encoder pull-up set in control word.

6. Flux Vector operation

Genus flux-vector uses the same core flux-vector algorithms as Gen4.

General setup procedures relevant to Gen4 are relevant to Genus (all differences are described in this document).

The Lm saturation is used in all calculations, - i.e. the same algorithms Gen4 code with the “Use saturation in slip and rotor time constant calculation (IM FV)” mode enabled.

As with Gen4, to understand the algorithms, the “Induction Motor Vector Data Template” spreadsheet can be used to assist in tuning. This spreadsheet will also produce fluxing maps to input into the controller.

N:\UK\Engineering\Documents\Motor Control\Vector Spreadsheets

The current control PI loops run at 2x the PWM frequency, so this is 16kHz for 8kHz PWM. This makes the system less sensitive to Kp and Ki gains than Gen4.

Current control PI gains are internally ‘real volts’, rather than ‘per unit’. This means that Gen4 PI gains are not directly portable to Genus. Use the PI gains of the starter DCF to begin tuning. A high-voltage motor setup (HVLP) may have considerably different gains than a low-voltage setup (Dragon8).

For example, a Kp of 1 may be typical for a Dragon8 setup, whereas 10 may be used in an HVLP at 700V.

7. Auto-tuning parameters on a dynamometer

Ensure that a valid starter .dcf is present and that some rotation is possible for a small torque demand (i.e. encoder sequence is correct).

If leakage inductances are known, enter them, otherwise put in the smallest possible value.

Ensure the new Genus debug items are viewable in the vehicle interface trace:

- 0x4602,52: User specified magnetizing inductance (Lm)
- 0x4602,53: Controller estimated magnetizing inductance (Lm)
- 0x4602,55: Controller estimated stator resistance (Rs)
- 0x4602,56: Controller estimated rotor resistance (Rr)

Im_min (4641,3) is traditionally set very low to keep some residual flux in the motor. It can now be real-time updated (sdo write to the object or ctrl+enter in the helper).

By changing it in real-time, we can see Lm for a range of Id currents and also see values for rr (when enabling auto-tune).

7.1. Magnetizing inductance

Set im_min to 10% of is_max, enable the controller and run the dyno to a speed where the modulation index is at least 10%. Ensure torque demand is zero.

Note the value of “0x4602,53: Controller estimated magnetizing inductance (Lm)”

Increase im_min over the range of id currents expected for the application (e.g up to 75% is_max) and note the values of Lm.

Keep an eye on the modulation index to ensure it never reaches 100% as the current level increases.

You should end up with a trace of Lm for different values of Id (Figure 2).

Do not forget that the stator leakage inductance is used in this calculation, so if this is not known, ensure it is set to a very low value to avoid corrupting the result.

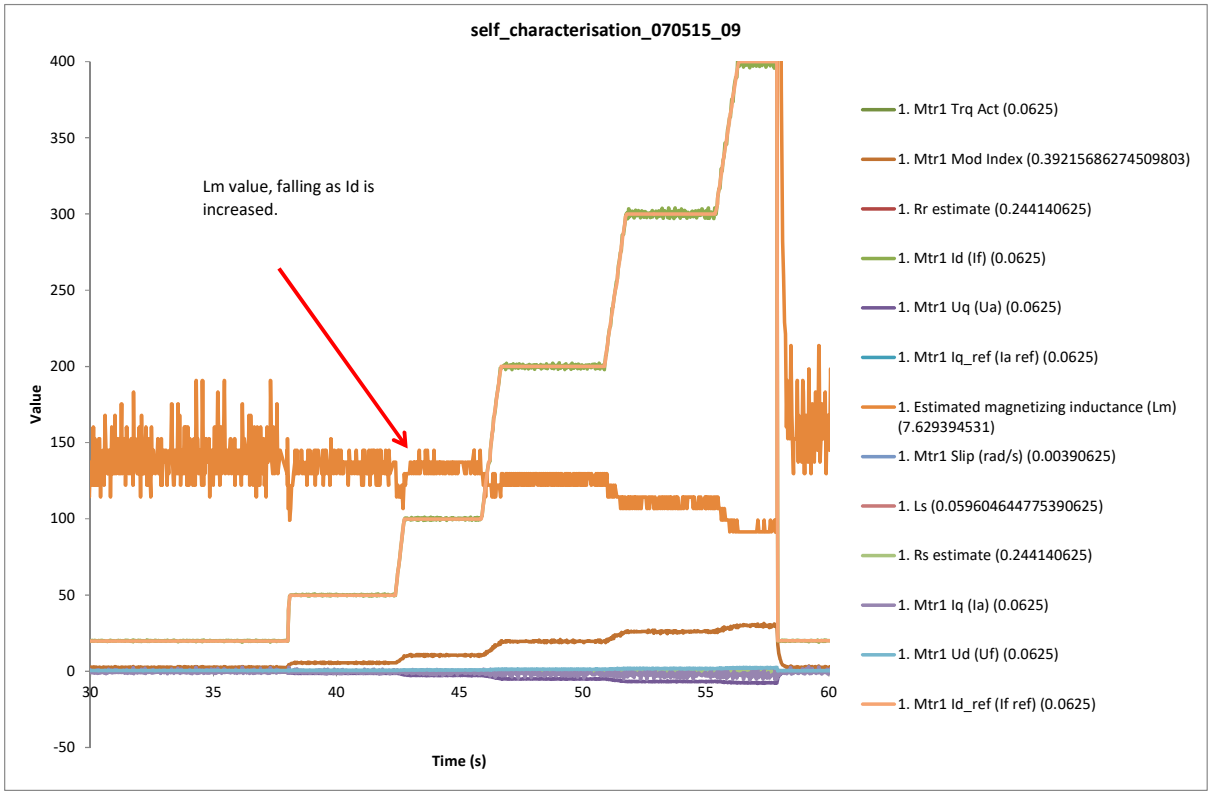


Figure 2: Lm saturation recording for a range of Id currents.

Note that:

$$L_m = L_s - L_{ls}$$

If you have a valid Lls (stator leakage inductance), then the Lm values shown can be used to help generate an Lm saturation table (0x4614). Remember that values will have to be converted into per-unit, with the nominal 1PU value input into 0x4641,9. The 1PU value can be anything with the Lm range, although it is traditionally set to a value for a low Id current.

If you don't have valid leakage inductances, then you need to make a guess at the leakage inductance and remove that from the Lm results (typically subtract 10% from each point).

If you wish to improve accuracy, then you can re-run the test for lower Id currents at higher speeds to try to keep the modulation index between 50 and 80% for all current points. The higher the modulation index, the more accurate the Lm result.

Once the Lm values are populated in the controller, you can compare them to the real-time measured values of Lm by looking at

- 0x4602,52: User specified magnetizing inductance (Lm)

This is useful for checking you've populated the values correctly.

Note that the controller estimated magnetizing inductance will attempt to calculate all the time, so please accept that the accuracy of the result will suffer when torque is applied or speed/modulation index is low.

7.2. Rotor and Stator Resistances

To attain a reasonable estimate of the rotor resistance, two things are necessary:

- An Id current level which does not result in significant saturation of Lm.
- Sufficient modulation index at the Id current level.

Typically an Id current of 25% of the maximum stator current will not result in a large saturation of Lm, so this is a sensible starting point (e.g. 100A on a 400A motor).

Ideally, with this Id current, the test speed of the dyno should result in 50% modulation index. This is often not possible, but at least 10% modulation index must be present – the higher the value, the better the result.

With the controller running and the Id current present, set the controller to pulse-test mode:

- sdo_wnx 1 0x4653 0 0x0020

The controller will now switch the Id pulse on and off every 4 seconds. No torque will be output until a key-recycle.

After each cycle, the Lm, Rr and Rs values will be updated. The value of Lm will be the same as that calculated via the method in section 7.1.

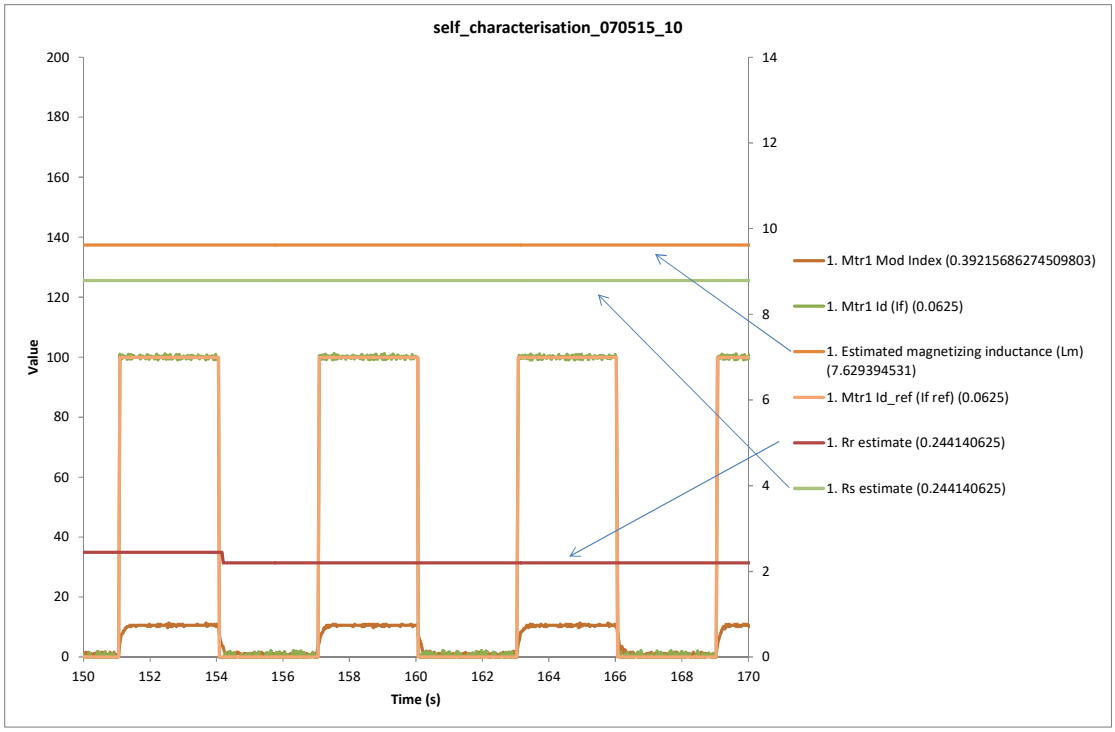


Figure 3: Pulse testing to measure Lm, Rr and Rs.

The user should analyse rotor resistance over a period of time and measure for a selection of low Id currents to ensure results are consistent.

The stator resistance measurement result may vary far more than the rotor resistance; however the accuracy is far less significant to the motor control algorithms. Where stator resistance is important is in the generation of a fluxing map via the spreadsheet (page 8) as this will tailor the fluxing for maximum efficiency based on this value. The engineer should consider the cable resistance as this will add significantly to this parameter – any dyno setup should be vaguely representative of the target vehicle – i.e. not a 10 metre cable run.

8. Appendix – Useful Maths

At steady state, the Ud and Uq voltages are:

$$U_q = R_s I_{q_ref} + (\omega_e + slip) L_s I_{d_ref}$$

$$U_d = R_s I_{d_ref} - (\omega_e + slip) l'_s I_{q_ref}$$

Ud and Uq values in the application should be close to those predicted by the Excel characterisation spreadsheet.

When torque is zero, Iq and slip are zero. So when we set an Id_ref current in a spinning motor, we can calculate a value of Ls from:

$$L_s = \frac{U_q}{\omega_e I_{d_ref}}$$

The controller needs Lm for its maths, so to get this we have to subtract the stator leakage inductance:

$$L_m = L_s - L_{ls}$$

Typically we've assumed the leakage inductance to be about 10% of the Lm inductance at a low-current test-point (i.e. when it's not saturating). However this isn't always the case and if the customer knows a better value, we should consider this.

Slip frequency is given by:

$$\omega_{slip} = \frac{R_r}{L_r} \frac{I_q}{I_d}$$

This basically means the higher the Iq to Id ratio, the bigger the slip.

The Iq to Id ratio is set by ls'. So if you put in 4 for this, then Iq can never by more than 4x Id. This is a good way to limit the slip – particularly in field weakening where the controller will force down Id to stay in control and increase Iq to maintain the torque.

If no value of Ls' is set, then the controller internally calculates a value based on Lm, Llr and Lls. You can see if this is inappropriate if the slip is flying off to huge values when under field weakening or with an inappropriate flux map (insufficient Id current).